ALEJANDRO ORTIZ PERDOMO

AI Engineer | Machine Learning Engineer

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alejortizp.github.io/curriculum_HV

PROFESSIONAL SUMMARY

Machine Learning Engineer with end-to-end experience from experimentation to production. Specialized in predictive models (**churn, propensity, NBO**) with quantifiable impact: **Lift 3.0**, **+15%** conversion, **+60%** contactability. Proficient in **Scikit-learn, LightGBM, PyTorch** and **Databricks + MLflow** for MLOps. Extensive **deep learning** experience with Transformers and NLP. Combines strong mathematical foundation (Statistics) with modern software engineering to build reliable and scalable ML pipelines.

WORK EXPERIENCE

AI Engineer — AI Practice

Jun 2025 – Present

Periferia IT Group

- **AI Agent Platform (SRPA) for the SDLC:** Technical design, conceptualization and documentation of a proprietary *Single-Responsibility Plug-and-play* agent ecosystem that automates the end-to-end development lifecycle (onboarding, user-story generation, IDE assistance, multi-language unit testing with ≥**85%** coverage, QA, documentation and technical assistants on LLMs such as **DeepSeek R-1** on **Azure AI Foundry**). Multi-cloud platform (**Azure / AWS / GCP**) used both internally and commercially, delivering **−70%** ramp-up and **60–70%** savings across the development cycle.
- **CV-Job Matching System:** Pipeline with **Azure Document Intelligence** + **AI Search** processing 500+ CVs/day with **95%** extraction accuracy and vector semantic search reducing screening time by **70%**. Running in production at **10k+ searches/month** with <200ms latency.
- **Pre-sales & Solution Architecture:** Build technical estimates, team sizing and reference architectures for **GenAI, Agentic AI and RAG** solutions for enterprise clients (banking, insurance, retail, energy and telco). Lead

EDUCATION

Master in Software Engineering
Universidad de los Andes
2025 – In Progress

B.S. in Statistics
Universidad del Valle
2011 – 2016

Data Science Diploma (DS4A)
Correlation One, 2022

Deep Learning Diploma
Univ. Autónoma Occidente, 2023

LANGUAGES

Spanish
Native

English
Professional Proficiency (B2)

KEY SKILLS

Effective Communication
Problem Solving
Analytical Thinking
Team Collaboration
Adaptability
Continuous Learning
Time Management
Technical Leadership

technical discovery with stakeholders and position the proprietary agent portfolio in commercial technical proposals.

Freelance AI Engineer — Dental Clinic
Jul – Aug 2026

Freelance

- **WhatsApp AI Agents:** Designed and built, as sole engineer, two agents with **LangGraph** + **Azure OpenAI**: appointment scheduling synced with **Outlook (Microsoft Graph)**, with LLM intent classification and escalation of complex cases; and clinical-category reminders with *human-in-the-loop* approval.
- **Deterministic guardrails & serverless deployment:** The LLM only interprets; booking is executed by deterministic code with locking to prevent double booking. Deployed on **Azure Functions** (queues with poison-queue handling and deduplication), Table Storage, **Key Vault** with managed identity and Application Insights alerts; **WhatsApp Cloud API** webhook with HMAC verification. Test-driven development with **118 tests**.

Machine Learning Engineer
Sep 2023 – Jun 2025

Global Hitss

- **GenAI & RAG:** Designed and deployed conversational agent with RAG that reduced support tickets by **40%**, processing **10k+ queries/month** with 85% resolution without escalation. Stack: LangChain, Azure OpenAI, FAISS.
- **MLOps:** Implemented CI/CD pipeline in **Databricks + MLflow** reducing deployment time from 2 weeks to **2 days**. Production models with automatic drift monitoring and retraining.
- **Reinforcement Learning:** Productized *Next Best Offer* (NBO) system with A/B testing that increased conversion by **15%** and revenue by **$10k per person**.
- **Churn Model:** Developed predictive model with **Lift of 3.0** and 82% recall in top decile, integrated to CRM for proactive retention. Impacted **50k+ customers**.

Machine Learning Consultant
Aug – Sep 2023

Grupo CINTE

- **ML Observability:** Designed comprehensive monitoring methodology for *data/model drift* detection with automated retraining alerts, adopted as the data science team's standard.
- **Optimization:** Audited and optimized LightGBM and Random Forest algorithms, improving computational efficiency and reducing inference times.

CERTIFICATIONS

AI-102: Microsoft Certified Azure AI Engineer Associate
Microsoft, Feb 2026

NLP Specialization (RAG/Transformers)
Coursera, Jun 2025

PyTorch Ultimate
Coursera, Mar 2025

MLOps and Databricks
Netec, Dec 2024

AWS Cloud Practitioner
Udemy, Feb 2022

TECHNICAL SKILLS

(years exp.)

GENAI & LLMS

LangChain 2+ (Agents, RAG) | **LangGraph** 1+
Azure OpenAI 2+ (GPT-4, Embeddings)
Prompt Engineering 2+
Vector DBs 2+ (FAISS, Qdrant, Pinecone)
LLM Evaluation 1+ (RAGAS)

AZURE AI & CLOUD

Azure AI Search 1+
Document Intelligence 1+
Azure AI Foundry 1+ | **Azure Functions** 1+
Databricks + MLflow 3+ (Workflows, Model Registry)
Delta Lake 1+ | **Unity Catalog** 1+

GOOGLE CLOUD PLATFORM (GCP)

Vertex AI 1+ (Training, Endpoints, Pipelines)
Cloud Run 1+ | **Pub/Sub** 1+
Compute Engine 1+ (MIG zonales)
Cloud Storage 1+ | **Cloud SQL** 1+

AMAZON WEB SERVICES (AWS)

EKS 1+ | **Lambda** 1+ | **DynamoDB** 1+
SNS 1+ | **SQS** 1+ | **RDS** 1+

ML/DL FRAMEWORKS

PyTorch 2+ | **TensorFlow** 3+
Scikit-learn 4+ | **XGBoost** 4+ | **LightGBM** 4+
Hugging Face Transformers 2+

Data Scientist

Mar 2023 – Jul 2023

NAE

- **Contactability Model:** Developed **LightGBM** model that increased effective contact rate by **60%** (from 25% to 40%), optimizing time windows and channels. Impacted campaigns with **100k+ contacts/month**.

Data Scientist II

May 2022 – Jan 2023

Xolit

- **Payment Propensity Model:** Developed predictive model for payment probability and portfolio agreement, maximizing collection through **automated management priorities** based on propensity scores.
- **Recommendation System:** Implemented optimal contact time-slot recommendation engine per account holder, increasing debt collection management effectiveness.
- **Customer Segmentation:** Designed clustering-based marketing strategy for account holders, enabling personalized communication per segment to optimize collection rates.
- **Monitoring:** Built **Power BI** dashboards for tracking collection KPIs and production model performance metrics.

OPEN SOURCE PROJECTS

🏦 RAG Assistant with Web Scraping and RAGAS Evaluation

— github.com/alejortizp/rag-banco

Conversational assistant over a bank's public website: robots.txt-compliant crawler, **Qdrant** indexing, cross-encoder reranking and per-session memory. Evaluated with **RAGAS** (50 questions, independent LLM judge): faithfulness **0.87**; the reranker improved context utilization by **+8 pts** (A/B).

Stack: Qdrant, Sentence-Transformers, Cross-Encoder, Groq Llama 3.1, RAGAS, Streamlit, Docker

🤖 Email Agent with LangChain

— github.com/alejortizp/email-agent

Intelligent agent that queries emails via IMAP, classifies with LLM, and generates contextual automatic responses. Reduces response time by **80%**.

Stack: LangChain, Azure OpenAI, IMAP, Python, Prompt Engineering

💬 NL-to-SQL/Pandas Agent

— github.com/alejortizp/nl-to-code

Agent that translates natural language to SQL queries and Pandas operations. Includes code validation, secure sandbox execution, and automatic results visualization.

Stack: LangChain, SQL, Pandas, Code Interpreter, ReAct Pattern

MLOPS & ENGINEERING

Python 7+ | **SQL** 7+ | **PySpark** 3+
Docker 2+ | **CI/CD** 2+
Apache Airflow
Kubernetes 1+ | **Terraform** 1+
Go 1+ | **Angular** 1+ | **Kotlin** 1+
FastAPI 3+ | **Streamlit** 3+
Git 4+ | **GitHub Actions** 2+

ACADEMIC PROJECTS — M.SC. (UNIANDES)

☁️ Teaching-Assistant Platform on GCP (Go + Angular)

MISW4204 Cloud Solutions Development · 2026

Top contributor (**60%** of commits, team of 4). Migrated the system from Docker Compose to an **autoscaling MIG + Load Balancer** and then to serverless **Cloud Run + Pub/Sub**, with asynchronous AI-generated PDF reports. JMeter load tests: **158 rps** (p95 292 ms) with **0%** infrastructure errors; **84%** (Go) and **93%** (Angular) coverage; SonarQube A rating.

Stack: Go (Gin), Angular, PostgreSQL, Cloud Run, Pub/Sub, Cloud SQL, OpenAI, JMeter, SonarQube, GitHub Actions

🚚 Event-Driven Delivery Marketplace on AWS

MISW4301 Cloud-Native App Development · 2026

Led the AWS architecture and IaC for **11 FastAPI microservices** (hexagonal architecture) on **EKS**, with a serverless extension: **SNS→SQS** with DLQs, **4 container Lambdas** and DynamoDB, all in **Terraform** deployed via GitHub Actions to 5 environments. Fixed cold starts and IAM permissions, reaching **76/76** E2E assertions with zero failures.

Stack: AWS (EKS, Lambda, SNS, SQS, DynamoDB, RDS), Terraform, Kubernetes, FastAPI, Docker

🛡️ Microservices Architecture Experiments

MISW4202 Agile Software Architectures · 2026

Hypothesis-driven experiments for a booking system: asynchronous payment-anomaly detection in **538 ms** mean vs. a 2 s target (t-test, $p<0.001$) and **100%** Man-in-the-Middle attack detection with zero false positives (RS256 JWT with revocation, RSA-OAEP, integrity hashing, TLS).

Stack: Flask, Redis Streams, Celery, Nginx, JWT, RSA, SciPy

⏰ QR Alarms: from UX Research to Android + Web App

— github.com/a-ortizp/alarmas-qr-app

UX Prototyping · MISO · 2026

Full product cycle: user research, **3 usability tests** (20 findings), Figma prototype and Material 3 design system (48 components); then a **22-screen** Android app (ML Kit QR scanning, alarms firing with the app closed) and Angular web app, with signed APK releases via GitHub Actions.

Stack: Figma, Kotlin, Jetpack Compose, CameraX, ML Kit, Angular, GitHub Actions